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IMPLEMENT TETRIS ON FPGA

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Abstract

- ❑ Tetris is a legendary blockstacking game that has become an icon in the history of video games. In this project, I have chosen to recreate Tetris on the FPGA DE2-115 kit, with display resolution of 640×480 @ 60Hz and using the PS/2 keyboard input device.



Motivation and Objectives

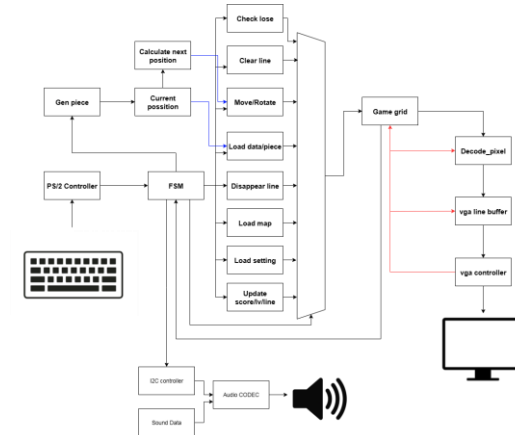
- ❑ Motivations
 - Even after more than three decades since its release, Tetris remains one of the most beloved and iconic puzzle games.
 - Classic Tetris World Championship (CTWC), where players such as have achieved legendary status.
- ❑ Objectives
 - Implement the entire Tetris game logic — including block generation, movement, collision detection, and scoring
 - Ensure the game runs smoothly and responsively with accurate timing control using 50 MHz clock.

Current result and future develop

- ❑ Result: The game run successfully at clock 50 MHz without any glitch found in the system yet.
- ❑ Develop: Mode B, C is in development state with the ambitious to create PvP/PvE mode with Tetris AI Bot

Hardware Design

- ❑ The process begins with the keyboard, connected through a PS/2 controller, which moving or rotating a Tetromino. These signals are then processed by the game logic, consisting of modules for piece generation, movement, rotation, collision detection, and line clearing.



Hardware Implementation

- ❑ The keyboard/speaker/VGA(with monitor) being connect to FPGA.



References

- [1] https://www.retrogames.cz/play_1030-NES.php
- [2] https://course.ece.cmu.edu/~ece545/F11/reports/F07_NetworkedTetris.pdf
- [3] https://tetris.wiki/Super_Rotation_System